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Endometriosis

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Overview

Background

Endometriosis is defined as the presence of normal endometrial mucosa (glands and stroma) abnormally implanted in locations other than the uterine cavity (see the image below). This tissue, possessing the same steroid receptors as normal endometrium, is capable of responding to the normal cyclic hormonal milieu. Microscopic internal bleeding, with the subsequent inflammatory response, neovascularization, and fibrosis formation, is responsible for the clinical consequences of this disease.[1]

This condition is a common, poorly understood, and extremely debilitating benign gynecologic condition. The psychological impact of the severe pain experienced by the patient is compounded by the negative impact of the disease on fertility.[2]

In the typical patient, the ectopic implants are located in the pelvis (ovaries, fallopian tubes, vagina, cervix, or uterosacral ligaments or in the rectovaginal septum) and manifest as severe dysmenorrhea, chronic pelvic pain, or infertility. More unusual implantation sites (laparotomy scars, pleura, lung, diaphragm, kidney, spleen, gallbladder, nasal mucosa, spinal canal, stomach, breast) can be responsible for bizarre symptoms such as cyclic hemoptysis and catamenial seizures.

The etiology and pathophysiology of endometriosis are not well understood. No cure exists for the disease, although the hormonal responsiveness of the implants can be exploited and provides the rationale for current methods of medical therapy. Treatment is directed toward medical suppression, surgical excision, and symptom alleviation.

Adenomyosis is the invasion of the myometrium by endometrial tissue.

The following are some images of active endometriosis and old scarring.

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Etiology

Several theories exist that attempt to explain this disease, although none have been entirely proven. Leading theories include metaplastic conversion of coelomic epithelium and hematogenous or lymphatic dispersion of endometrial cells. It is likely a combination of various factors that cause and determine the severity of this disease.

Previous theories suggest that endometriosis results from the transport of viable endometrial cells through retrograde menstruation. Cells flow backward through the fallopian tubes and deposit on the pelvic organs, where they seed and grow. A population of cells that retain stem cell properties resides in the endometrium. It may be these properties that allow these cells to survive in ectopic locations.

Risk factors

Risk factors for endometriosis include the following:

- Family history of endometriosis
- Early age of menarche
- Short menstrual cycles (< 27 d)

- Long duration of menstrual flow (>7 d)
- Heavy bleeding during menses
- Inverse relationship to parity
- Delayed childbearing
- Defects in the uterus or fallopian tubes
- Hypoxia and iron deficiency may contribute to the early onset of endometriosis

Retrograde menstruation

Early in the 20th century (1927), Samson proposed his theory of retrograde menstruation through the fallopian tubes into the peritoneal cavity as a cause of endometriosis. Subsequent studies have shown that retrograde menstruation is a common physiologic event and cannot adequately explain the extrauterine implantation of endometrial tissue. Diagnostic laparoscopy during the perimenstrual period has shown that as many as 90% of women with patent fallopian tubes have bloody peritoneal fluid.

Nonetheless, conditions that increase the rate of retrograde menstruation, such as congenital outflow tract obstructions, do increase the risk of endometriosis. Various animal experiments and clinical observations support this theory.[3,4]

Because most women do not have endometriosis, perhaps immunologic or hormonal dysfunction leaves some women predisposed.

Immunologic dysfunction

Research has suggested involvement of the immune system in the pathogenesis of endometriosis. An altered immune response to the displaced endometrial tissue has been shown to play an important role as well. Women with this disorder appear to exhibit increased humoral immune responsiveness and macrophage activation while showing diminished cell-mediated immunity with decreased T-cell and natural killer cell responsiveness. Humoral antibodies to endometrial tissue have also been found in sera of women with endometriosis.[5]

Intriguing nonhuman primate studies have demonstrated a strong association between dioxin exposure and the development of endometriosis, implying further that dysfunction of the immune system may contribute to this disease.[6] Epidemiologic investigations have supported but have not been able to confirm this association in humans.[7]

Studies have assessed the differences between eutopic endometrium and endometriosis. In endometriosis, an aberrantly expressed factor SF-1 activates the expression of the enzyme aromatase, which converts C19 steroids to estrogens. Consequently, estrogen increases the synthesis of prostaglandin E2, which exerts a positive feedback effect, resulting in increased aromatase activity. Additionally, endometriotic tissue is deficient in the enzyme 17-beta hydroxy steroid dehydrogenase type 2, which converts E2 in eutopic endometrium to the less potent E1 under the direction of progestins.

One study found a higher number of endometriomas, more bilateral disease, and a higher incidence of significant pain in women with aromatase positive disease.[8] However, other studies have shown increased cyclooxygenase-2 (COX-2) expression in the stromal cells and aberrant aromatase expression in eutopic endometrium of women with endometriosis.[9,10]

Successful treatment has been described with aromatase inhibitor therapy in women with severe postmenopausal endometriosis,[11] and studies have also shown it to be effective for severe endometriosis in premenopausal women. Aromatase inhibitors are a promising treatment option for endometriosis-associated pain but should be used in combination with progestins in premenopausal women to avoid ovarian stimulation and mitigate bone loss.[12]

Metaplasia

Metaplasia, or the changing from one normal type of tissue to another normal type of tissue, is another theory. The endometrium and the peritoneum are derivatives of the same coelomic wall epithelium.

Transformation of coelomic epithelium into endometrial-type glands in response to as yet unknown stimuli could explain endometriosis in unusual sites.[13] Coelomic metaplasia is also believed to explain the occurrence of endometriosis in women who have undergone total hysterectomy and are not taking estrogen replacement.[14] Peritoneal mesothelium has been postulated to retain its embryologic ability to transform into reproductive tissue. Such transformation may occur spontaneously, or it may be facilitated by exposure to chronic irritation by retrograde menstrual fluid.

Remnant Mullerian cells

Another theory states that remnant Mullerian cells may remain in the pelvic tissues during development of the Mullerian system. Under situations of estrogen stimulation, they may be induced to differentiate into functioning endometrial glands and stroma.

Genetics

Some women may have a genetic predisposition to endometriosis. Studies have shown that first-degree relatives of women with this disease are more likely to develop it as well.

Anatomic spread, dissemination and deposition

Transtubal dissemination is the most common route, although other routes, such as lymphatic and vascular channels, have been observed. This may explain how endometrial tissue can be found at distant, noncontiguous locations in the body.

Iatrogenic deposition of endometrial tissue has been found in some cases following gynecologic procedures and cesarean sections.

The ovary is the most common site for endometriosis. Spread to the ovary is believed to be lymphatic,[15] although superficial implants may be due to retrograde menstrual flow, because the ovaries are in a dependent part of the pelvis.

Lesions can vary in size from spots to large endometriomas. The classic lesion is a chocolate cyst of the ovary that contains old blood that has undergone hemolysis (see the first image below). Once intracystic pressure rises, the cyst perforates, spilling its contents within the peritoneal cavity. This can cause the severe abdominal pain typically associated with endometriosis exacerbations. The inflammatory response causes adhesions that further increase the morbidity of the disease (see the second image below).

Uterine serosa can be affected. Vesicular lesions may provoke an inflammatory response and scarring that cause the bladder to adhere anteriorly. Posteriorly, the disease may cause obliteration of the cul-de-sac and form dense adhesions between the posterior vaginal wall or cervix and the anterior rectum. Severe dyspareunia, dyschezia, and alteration of bowel habits are the clinical sequelae of this common spread.

Deep peritoneal disease is caused by infiltration of the uterosacral ligaments and rectovaginal septum by endometriotic nodules. Tethering of the uterus can lead to fixed retroversion. Dyspareunia is an important feature.

Through contiguous spreading, endometriosis may invade the rectovaginal septum and the anterior rectal wall. It may also involve the upper rectum and sigmoid colon, infiltrating the muscularis. Cyclical rectal bleeding (hematochezia) is pathognomonic of endometriosis. However, transmural bowel involvement by endometriosis remains a rarity. The ileum, appendix, and cecum may also be involved, leading to intestinal obstruction. Cicatrization as a consequence of endometriosis may lead to symptoms of obstruction even in postmenopausal women.

Although uncommon, interference in the genitourinary tract by endometriosis can affect the bladder, ureters, and kidneys by invasion, compression, or scarring. Medical therapy has less than satisfactory results, and surgical intervention is often required.

Uncommon sites include incisional scars, the umbilicus, and the thoracic cavity. Catamenial or cyclic pneumothorax can cause hemoptysis.

Ectopic endometrial tissue theoretically can undergo malignant transformation; histologic evaluation may be necessary. Significant predictive factors for the malignant transformation of endometriosis appear to include age older than 49 years and cysts that are multilocular and have solid components.[16] Although elevated, levels of serum cancer antigen 125 (CA-125) do not appear to be a significant predictor of malignant transformation of endometriosis.

Postmenopausal endometriosis may be encountered in women who are on estrogen therapy. Occasionally, if estrogen therapy is administered after total abdominal hysterectomy, endometriosis can be stimulated in an ovarian remnant. Extrapelvic endometriosis is believed to be hormone-resistant when it occurs after surgical castration. Transplantation of endometrial implants during the original surgery is believed to explain this occurrence. Another possible explanation is coelomic metaplasia.

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Pathophysiology

Ectopic endometrial tissues are most commonly located in the dependent portions of the female pelvis (eg, posterior and anterior cul-de-sac, uterosacral ligaments, tubes, ovaries), but any organ system is potentially at risk (see the following images).

These ectopic foci respond to cyclic hormonal fluctuations in much the same way as intrauterine endometrium, with proliferation, secretory activity, and cyclic sloughing of menstrual material. The products of this metabolic activity, including the concentrated and cyclic release of cytokines and prostaglandins, lead to an altered inflammatory response characterized by neovascularization and fibrosis formation. Some investigators have been able to demonstrate abnormal T- and B-cell function, abnormal complement deposition, and altered interleukin (IL)-6 production in women with this disease.

The associated pain, adhesion formation (see the image below), and anatomic distortion are responsible for the clinical consequences of this disease.

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Epidemiology

Endometriosis occurs in 6-10% of US women in the general population, and approximately 4 per 1000 women are hospitalized with this condition each year.[17,18] The exact incidence in the general population is unknown, because the definitive diagnosis requires biopsy or visualization of the endometriotic implants at laparoscopy or laparotomy. Most prevalence studies are based on a surgical population in which the likelihood of disease is greater. The best estimates of incidence in the general population come from women with proven fertility undergoing tubal sterilization procedures.

Endometriosis is an estrogen-dependent disease and, thus, usually affects reproductive-aged women. This condition has a prevalence rate of 20-50% in infertile women, but it can be as high as 71-87% in women with chronic pelvic pain.[17,19] In a large series involving adolescent females with chronic pelvic pain, 45% were found to have endometriosis at laparoscopy. Of note, only 25% had a normal pelvis. In that series, the rate of endometriosis was found to increase with age from 12% in females aged 11-13 years to 45% in females aged 20-21 years.[20] In an earlier study, evidence of endometriosis was found during laparoscopy in 20-50% of asymptomatic women.[21]

A systematic review and meta-analysis revealed a high prevalence of endometriosis among adolescent females with pelvic pain. Of 1011 symptomatic adolescents who underwent laparoscopic investigation, 648 (64%) were found to have endometriosis.[22]

A familial association exists, with a 10-fold increased incidence in women with an affected first-degree relative.[23] Monozygotic twins are markedly concordant for endometriosis.[24]

Race-, sex-, and age-related demographics

Most research and case studies have been performed in White populations; however, no difference appears to exist among ethnic or social groups.[25] Although endometriosis is obviously a disease largely confined to the female population, interestingly, scattered case reports exist of lesions that are histologically indistinguishable from endometriosis found in men exposed to high-dose exogenous estrogens.

Endometriosis is largely confined to women of reproductive age with an active hypothalamic-pituitary-ovarian axis. Pelvic endometriosis typically occurs in women aged 25-30 years. Extrapelvic manifestations of this disorder occur in woman aged 35-40 years. Women younger than 20 years with this disease often have anomalies of the reproductive system. Prepubertal girls do not seem to be at risk for this disease, although the number of reports of endometriosis in young women shortly after menarche is increasing.

Menopause (whether spontaneous or induced through surgical or medical means) usually leads to resolution of symptoms. The disease seems to remain quiescent even in patients receiving menopausal hormone therapy.

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Prognosis

Endometriosis has been found to resolve spontaneously in one third of women who are not actively treated.[26] However, it is generally a progressive disease, with an unpredictable extent of progression and subsequent morbidity. Although most patients (up to 95% in some studies) respond to medical therapy (suppression of ovulation) for decreasing pelvic pain, such therapy is ineffective for treatment of endometriosis-associated infertility but does preserve the potential for conception. Nonetheless, as many as 50% of women have a return of symptoms within 5 years with medical management.

Combination estrogens/progestins relieve pain in as many as 80-85% of patients with endometriosis-related pelvic pain. After 6 months of danazol therapy, as many as 90% of patients with moderate endometriosis experience adequate pain relief.

Findings from the prospective Australian Longitudinal Study on Women's Health of 9585 women (age at trial onset, 18-23 y) indicated that, regardless of fertility status, previous exposure to oral contraceptive pills (OCPs) reduced the risk of a subsequent diagnosis of endometriosis among parous women but increased the risk among nulliparous women.[27]

Minimally invasive surgical therapy improves fertility rates, but this treatment is not as effective at eliminating pain. Definitive surgical therapy (total hysterectomy with bilateral salpingo-oophorectomy [TAH-BSO] and peritoneal stripping) offers the greatest chance for long-term resolution of pain (up to 90%). However, reserve this option as a last resort in patients with completely incapacitating disability or those who have no desire for future childbearing.

In general, pregnancy is possible but depends on the severity of the disease. Endometriosis signs and symptoms generally regress with the onset of menopause and during pregnancy.

Endometriosis has been linked to an increased risk of ovarian cancer. The results of a population-based cohort study showed that women with a history of endometriosis have a fourfold increased risk for the development of ovarian cancer. Women with deep infiltrating endometriosis and/or ovarian endometriomas have a 9.7-fold higher risk of ovarian cancer compared with individuals who do not have endometriosis.[28]

Mortality/morbidity

Acute or chronic pelvic pain is common in patients with endometriosis. Infertility is also common. Approximately 30-40% of women with endometriosis will be subfertile.

Adolescent patients typically present with increasingly severe dysmenorrhea and/or chronic pelvic pain. Any persistent symptoms that seem cyclic in nature should prompt the practitioner to consider a search for endometriosis.[29]

Symptoms do not correlate well with disease severity. Significant dysfunction can be present with minimal gross disease, whereas severe endometriosis is sometimes asymptomatic. The pain of endometriosis responds poorly to antiprostaglandins and OCPs. Symptoms are related to the site of endometriotic implants and the organ system involved.

Cases have been reported of extrapelvic involvement in virtually every other organ system including the central nervous system (CNS), lungs, pleura, kidney, and bladder.[30] The gastrointestinal (GI) tract is the most common extrapelvic site of endometriosis, and symptoms include bowel obstruction, rectal bleeding, and constipation. Symptoms in other locations are related to the site and size of endometrial implants.

Mortality is negligible.

Complications

Complications of endometriosis may include or fall into the following three categories:

- Infertility/subfertility
- Chronic pelvic pain and subsequent disability
- Anatomic disruption of involved organ systems (eg, adhesions, ruptured cysts)

Endometriosis has also been linked to adverse pregnancy outcomes. Farland et al found that pregnant women with a history of endometriosis were at increased risk for spontaneous abortion (relative risk [RR], 1.40), ectopic pregnancy (RR, 1.46), gestational diabetes (RR, 1.35), and hypertensive disorders (RR, 1.30).[31]

A genetic association study showed that endometriosis is linked to increased risk of depression, anxiety disorders, and eating disorders.[32]

In a population-based cohort study, patients with endometriosis had a 42% higher risk for self-harm, overdose, or suicide compared with matched controls. The risk was greatest among patients with minimal or no previous mental health services utilization.[33]

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Presentation

History

Common elements in the history include nulliparity and regular menstrual cycles with prolonged flow of 8 or more days. Onset of pain usually precedes flow by a few days and begins to resolve 1-2 days into the menses. Symptoms also usually improve during pregnancy and after menopause; they can recur postpartum or with postmenopausal hormone therapy.

A familial/genetic predisposition has been documented. A woman with a first-degree relative with endometriosis has a lifetime risk of the disease approximately 10 times that of a woman without an affected family member.

When the products of cyclic sloughing of endometriotic implants become entrapped by cyst formation, the resulting mass is referred to as an endometrioma. These can occur in any location but are most commonly found involving one or both ovaries. These masses can become quite painful, and patients with rupture present with an acute surgical abdomen.

Symptoms

Although a significant number of women with endometriosis remain asymptomatic (approximately one third),[34] the most important point to remember is that the degree of visible endometriosis has no correlation with the degree of pain or other symptomatic impairment, because the location and depth of endometrial implants affect the symptomatology. However, pain does correlate with the depth of tissue infiltration, as pain is thought to be related to the degree of peritoneal inflammation rather than the volume of implants.

Associated intrapelvic/intra-abdominal adhesions are also important determinants of the degree of pain experienced. Midline disease is generally believed to be more painful than lateral disease. In addition to pain, patients present with nonspecific symptoms of fatigue, generalized malaise, and sleep disturbances.

Symptoms of endometriosis can be variable but typically reflect the area of involvement. Such symptoms may include the following:

- Dysmenorrhea
- Heavy or irregular bleeding
- Pelvic pain
- Lower abdominal or back pain
- Dyspareunia
- Dyschezia (pain on defecation) often with cycles of diarrhea and constipation
- Bloating, nausea, and vomiting
- Inguinal pain
- Pain on micturition and/or urinary frequency
- Pain during exercise

Because most endometriotic implants are found on the uterus, ovaries, and posterior peritoneum, the patient usually presents with a history of progressively increasing pelvic pain and/or secondary dysmenorrhea. Not uncommonly, women report painful bowel movements, diarrhea, or even hematochezia in association with their menses when endometriosis involves the rectosigmoid colon. Likewise, dysuria, flank pain, or hematuria may be present if the bladder or ureters are involved.

Cyclic pain is pain that accompanies bleeding at the time of menstruation. This could involve the bladder (hematuria), bowel (hematochezia and painful defecation), or, rarely, bleeding at uncommon sites such as the umbilicus, abdominal wall, or perineum. Occasionally, patients present with a cyclically painful expanding mass in a pelvic surgery scar; excision reveals a focus of endometriosis.

Acute exacerbations are believed to be caused by chemical peritonitis due to leakage of old blood from an endometriotic cyst. With conscious laparoscopic pain mapping, painful lesions were found to involve peripheral spinal nerves rather than autonomic nerves.[35]

Secondary dysmenorrhea occurs twice as often in women with endometriosis as in controls.[21] Pain frequently commences before menses. Endometriosis should be considered in a patient presenting with significant dysmenorrhea, and the patient should be started on empiric therapy.

Patients who are sexually active may report deep dyspareunia that is worse in the premenstrual phase of the cycle. Deep dyspareunia may be due to scarring of the uterosacral ligaments, nodularity of the rectovaginal septum, cul-de-sac obliteration, and/or uterine retroversion, all of which may also lead to chronic backache. These symptoms are exaggerated during menses. Women with deep infiltration of the uterosacral ligaments were shown to have the most severe impairment of sexual function.[36]

More uncommon cyclic symptoms include hemoptysis (pulmonary involvement), catamenial seizures (endometriotic lesions in the brain), and umbilical bleeding (implants in the umbilicus).

Partial or complete bowel obstruction occasionally occurs because of either adhesion formation or a circumferential endometriosis lesion. Ureteral obstruction and hydronephrosis can result from endometrial implants on the ureter or mass effect from an endometrioma.

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Physical Examination

Frequently, patients with endometriosis do not have any physical findings beyond tenderness related to the site of involvement. The most common finding is nonspecific pelvic tenderness.

On pelvic evaluation, tenderness upon examination is best detected at the time of menses. The hallmark finding on examination is the presence of tender nodular masses along thickened uterosacral ligaments, the posterior uterus, or the posterior cul-de-sac. Obliteration of the cul-de-sac in conjunction with fixed uterine retroversion implies extensive disease. Occasionally, a bluish nodule may be seen in the vagina due to infiltration from the posterior vaginal wall.

Rupture of an ovarian endometrioma may present as an acute abdomen.

Extensive involvement of the rectum and other areas of the gastrointestinal (GI) tract may cause adhesions and obstruction.

Examination should also include evaluation for cervicitis, abnormal discharge, and sexually transmitted infections (STIs).

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Staging

American Society for Reproductive Medicine classification

The revised American Society for Reproductive Medicine classification of endometriosis is the most widely used staging system.[37] Point scores are assigned based on the number of lesions and their bilaterality as well as associated adhesion formation noted at the time of surgery. Lesion size is also a scoring factor.

This classification is a fairly accurate method of recording laparoscopic findings. However, high intraobserver and interobserver variability precludes its use in comparing the outcomes of therapeutic studies. Furthermore, this staging system does not correlate well with pain and dyspareunia, and fecundity rates cannot be predicted accurately.

However, the patient's stage (ie, 1-4, or minimal, mild, moderate, and severe) may be useful in determining the prognosis for subsequent reproduction. The staging system can also be used to monitor a patient's response to therapeutic efforts. Surgical exploration is required for this staging system, both initially and for subsequent follow-up; a discussion of its details is beyond the scope of this article.

Enzian classification

The Enzian classification system was created to stage deep infiltrating endometriosis, and it was updated to include ovarian endometriosis, peritoneal disease, and adhesions. The system can be used for both preoperative imaging (magnetic resonance imaging and ultrasonography) and intraoperative findings.[38]

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Diagnostic Considerations

In 15% of cases of pelvic pain, endometriosis is the underlying cause. It should be considered in women with chronic pelvic pain who do not respond to standard nonsteroidal anti-inflammatory drugs (NSAIDs) or oral contraceptive therapy. This nonresponse to these agents is in contrast to the response among patients with primary dysmenorrhea.

Other conditions that should be considered when evaluating a patient with suspected endometriosis include the following:

- Adenomyosis
- Colon cancer
- Dysmenorrhea
- Functional or neoplastic ovarian cyst
- Ovarian cancer
- Pelvic adhesions
- Serositis
- Uterine malformation

Differential Diagnoses

- Appendicitis
- Chlamydia (Chlamydial Genitourinary Infections)
- Urinary Tract Infection (UTI) and Cystitis (Bladder Infection) in Females
- Diverticulitis
- Gonorrhea
- Ovarian Cysts
- Pelvic Inflammatory Disease
- Ovarian (Adnexal) Torsion
- Ectopic Pregnancy

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Workup

Workup

Approach Considerations

Few laboratory tests prove to be valuable in the diagnosis of endometriosis, although some tests may be helpful in ruling out specific conditions in the differential diagnosis.

Routine radiographs are not recommended unless other disease entities requiring these studies are in the differential diagnosis.

Transvaginal ultrasonography is the first-line imaging modality for suspected endometriosis.[17,39] Magnetic resonance imaging (MRI) is indicated as a second-line modality, particularly when ultrasound results are inconclusive or if deep endometriosis is suspected. Computed tomography (CT) has limited utility in pelvic endometriosis but may be used to detect complications.

Imaging results may be normal in patients with superficial peritoneal disease.[40]

Gross visualization of endometrial implants remains the definitive method of diagnosis. In this era of minimally invasive surgery, laparoscopy is the procedure of choice. Laparotomy can be another method of diagnosis. This is usually performed when another cause of patient pain is suspected.

Hysterosalpingography may reveal tubal occlusion or periadnexal adhesions.

Conscious pain mapping (ie, with the patient awake) has been used to locate the specific areas that cause pain.[35] Subsequently, the patient is placed under anesthesia and the deposits are ablated.

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Laboratory Studies

A complete blood cell (CBC) count with differential may help differentiate pelvic infection from endometriosis as well as assess the degree of blood loss.

Urinalysis and urine culture should be ordered if urinary tract infection (UTI) is in the differential diagnosis. In addition, cervical Gram stain and cultures should be considered, because sexually transmitted infections (STIs) can also cause pelvic pain and infertility.

Serial measurements of serum cancer antigen 125 (CA-125) have a low sensitivity in detecting endometriosis, although levels may be elevated in advanced cases. However, normal posttreatment values do not mean that endometriosis is absent. Thus, the test lacks adequate sensitivity or specificity to be of clinical value and is not recommended for the diagnosis of endometriosis.[39,40]

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Ultrasonography

The National Institute for Health and Care Excellence (NICE) recommends that transvaginal ultrasonography be offered to all patients with suspected endometriosis. If a transvaginal ultrasound scan is not appropriate, a transabdominal ultrasound scan of the pelvis can be considered as an alternative.[39] The European Society of Human Reproduction and Embryology (ESHRE) recommends either ultrasonography or MRI in the workup of endometriosis.[40]

The ultrasonographic features of endometriomas vary from simple cysts to complex cysts with internal echoes to solid masses, usually devoid of vascularity.[41]

Transvaginal ultrasonography is a useful method of identifying the classic chocolate cyst of the ovary. The typical appearance is that of a cyst containing low-level homogenous internal echoes consistent with old blood.

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MRI and CT Scanning

Magnetic resonance imaging (MRI) offers a superior combination of 3-dimensional (3-D) imaging with high-resolution special and temporal resolution, low observer dependency, no radiation exposure, and none of the risks associated with iodinated contrast agents.

With dynamic contrast-enhanced MRI, dynamic changes in MR signal intensity in selected tissues can be detected. Some of the newer generation contrast agents can be loaded with specific antibodies that allow for targeted imaging.

MRI is helpful in detecting rectal involvement and has been shown to accurately detect rectovaginal endometriosis and cul-de-sac obliteration in more than 90% of cases when ultrasonographic gel was inserted in the vagina and rectum.[42] MRI has a higher sensitivity for detecting pelvic masses than ultrasonography but is limited in identifying diffuse pelvic endometriosis.

On computed tomography (CT) scanning, endometriomas may appear as cystic masses, but their appearance is nonspecific and this imaging modality should not be relied on for diagnosis. Complications of endometriosis, including bowel obstruction and hydronephrosis, may be seen on CT scans.

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Laparoscopy and Biopsy

Laparoscopy is considered the primary diagnostic modality for endometriosis. This is an invasive procedure with an overall sensitivity of 97% and a specificity of only 77%.

Endometriosis has been described as protean in appearance. The classic lesions are blue-black or have a powder-burned appearance (see the first two images). However, the lesions can be red, white, or nonpigmented. Peritoneal defects and adhesions are also indicative (see the second two images). Bear in mind that microscopic evidence of endometriosis may be found in normal-appearing peritoneum.

The most common sites of involvement found during laparoscopy are the following, in descending order (see the following images):

- Ovaries
- Posterior cul-de-sac
- Broad ligament
- Uterosacral ligament
- Rectosigmoid colon
- Bladder
- Distal ureter

The following images are from various sites of endometriosis involvement.

In a study of low-grade ovarian endometrial adenocarcinomas, KRAS mutations were identified in 29% of endometriosis-associated tumors but only 3% in which endometriosis was not identified. This finding may be relevant to future targeted therapies.[43]

Histologic features

Histologic demonstration of both endometrial glands and stroma in biopsy specimens obtained from outside the uterine cavity is required to make the diagnosis of endometriosis. Occasionally, the finding of fibrosis in combination with hemosiderin-laden macrophages is sufficient for a presumptive diagnosis.

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Treatment

Approach Considerations

The dependence of endometriosis on the cyclic production of menstrual cycle hormones provides the basis for medical therapy. Medications used to treat endometriosis include oral contraceptive pills, progestins, androgens, and gonadotropin-releasing hormone (GnRH) agonists and antagonists. Each of these interrupts the normal cyclic production of reproductive hormones. Studies support the use of aromatase inhibitors for refractory or recurrent endometriosis.

Pain management

Options for the treatment of endometriosis-associated pain include analgesics, hormonal therapy, and surgery.[40] NICE recommends an initial short trial of acetaminophen or a nonsteroidal anti-inflammatory drug (NSAID).[39] However, a Cochrane review found no evidence of benefit for NSAIDs in placebo-controlled studies.[44] A meta-analysis of randomized, controlled trials by Samy et al showed that the options for hormonal therapy seem to be equally effective in relieving endometriosis-related pain.[45]

Endometriosis and subfertility

Peritubal and periovarian adhesions can interfere mechanically with ovum transport and contribute to subfertility. Peritoneal endometriosis has been postulated to contribute to subfertility by interfering with tubal motility, folliculogenesis, and corpus luteum function. Aromatase is believed to increase the prostaglandin E levels via increase in the cyclooxygenase-2 (COX-2) expression. Endometriosis may also cause subfertility by causing more sperm binding to the ampullary epithelium, thereby affecting sperm-endosalpingeal interactions.

Medical treatment of minimal or mild endometriosis has not been shown to increase pregnancy rates. Moderate to severe endometriosis should be treated surgically.

Other options for achieving pregnancy include intrauterine insemination, superovulation, and in vitro fertilization. In a case-controlled study, pregnancy rates with intracytoplasmic sperm injection were not affected by the presence or extent of endometriosis.[46] Other analyses have shown improvement with in vitro fertilization pregnancy rates with pretreatment of stage 3 and 4 endometriosis with gonadotropin-releasing hormone (GnRH) agonists.

Some authorities have recommended that endometriosis be suppressed prophylactically by continuous combined oral contraceptives, GnRH analogues, medroxyprogesterone, or danazol in order to cause regression of asymptomatic disease and enhance subsequent fertility. However, according to a Cochrane review, no benefit is derived from ovulation suppression in subfertile women with endometriosis who wish to conceive.[47]

Another Cochrane review found no evidence that interventions for the management of endometriomata before the use of assisted reproductive technology (ART) affected reproductive outcomes.[48]

In a systematic review and meta-analysis of the effect of endometriosis on ART outcomes from 36 studies, investigators found similar outcomes for live births between women with endometriosis who underwent in vitro fertilization and intracytoplasmic sperm injection and women without endometriosis.[49] However, women with severe endometriosis had lower live birth rates, clinical pregnancy rates, and mean number of retrieved oocyte relative to those without endometriosis. The investigators noted there is insufficient evidence to recommend surgery routinely before women undergo ART procedures.[49]

Surgical ablation of asymptomatic endometriosis has been shown to improve fecundity rates on a 3-year follow-up.[50]

The Endometriosis Fertility Index (EFI) is a validated tool for predicting non-ART pregnancy following endometriosis surgery. A meta-analysis of 17 studies (4598 women) showed that cumulative pregnancy rates at 36 months correlated well with the EFI score.[51]

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Hormonal Therapy

Hormonal therapy is the mainstay of treatment for endometriosis.[40] Combination oral contraceptive pills (COCPs), progestins, danazol, and gonadotropin-releasing hormone (GnRH) agonists and antagonists with or without hormone therapy are some of the medications used. Aromatase inhibitors are generally reserved for refractory or recurrent endometriosis.

Combination oral contraceptive pills

COCPs act by ovarian suppression and continuous progestin administration. Initially, a trial of continuous or cyclic COCPs should be administered for 3 months. With pain relief, this treatment is continued for 6-12 months. Pregnancy rates following discontinuation of the pill are 40-50%. This applies to a population unselected for stage and fertility status of the disease.

Continuous noncyclical administration of COCPs, omitting the placebo menstrual tablets, for 3-4 months helps avoid any menstruation and associated pain. A study by Guzick et al examined a head-head-to-head comparison of leuprolide and continuous oral contraceptives for the treatment of endometriotic pelvic pain; both were found to be equally effective.[52]

Women with endometriosis are at increased risk for epithelial ovarian cancer, and COCPs may reduce this risk.[53]

Progestational agents

All progestational agents act by decidualization and atrophy of the endometrium.

Medroxyprogesterone acetate has proven efficacy in pain suppression in both the oral and injectable depot preparations.[54,55] Oral doses of 10-20 mg/d can be administered continuously. The time to resumption of ovulation is longer and variable with depot preparations. Adverse effects include weight gain, fluid retention, depression, and breakthrough bleeding.

Megestrol acetate has been used in doses of 40 mg with similarly good results.[56]

The levonorgestrel intrauterine system (LNG-IUS) has been shown to reduce endometriosis-associated pain.[57] When inserted at the time of laparoscopic surgery, it has been found to reduce the recurrence of dysmenorrhea by 35%.[58]

Gonadotropin-releasing hormone therapies

Gonadotropin-releasing hormone analogues

GnRH analogues produce a hypogonadotrophic-hypogonadic state by downregulation of the pituitary gland. Goserelin and leuprolide acetate are commonly used agonists. Efficacy is limited to pain suppression, and fertility rates show no improvement.[59]

Studies indicate that GnRH agonist therapy may lead to improvement in endometriosis-associated pain in approximately 70-90% of patients.[60] However, pain relief is often transient; symptoms typically recur within 3 to 6 months following the cessation of treatment, particularly in patients with deep infiltrating endometriosis or when suppressive therapy is not followed by hormonal maintenance.

Treatment is usually restricted to monthly injections for 6 months. Loss of trabecular bone density caused by GnRH is restored by 2 years after cessation of therapy.[61] Other prominent adverse effects include hot flashes and vaginal dryness.

Gonadotropin-releasing hormone antagonists

Elagolix is an oral GnRH receptor antagonist that inhibits endogenous GnRH signaling by binding competitively to GnRH receptors in the pituitary gland. It was approved by the US Food and Drug Administration (FDA) for management of moderate to severe pain associated with endometriosis. Administration results in dose-dependent suppression of luteinizing hormone (LH) and follicle-stimulating hormone (FSH), leading to decreased blood concentrations of the ovarian sex hormones, estradiol and progesterone.

Approval of elagolix was based on the results of two clinical trials.[62] Improvement of dysmenorrhea was reported in up to 46% of women in the lower-dose elagolix groups and up to 76% in the higher-dose elagolix groups, compared with up to 23% in the placebo groups. Nonmenstrual pelvic pain improvement was reported by up to 50% of women in the lower-dose elagolix groups and up to 58% in the higher dose groups, compared with 37% in the placebo groups.

Elagolix is associated with hypoestrogenic adverse effects, including hot flushes, elevated serum lipids, and decreases from baseline in bone mineral density. Thus, treatment duration recommendations are specified for each dosage regimen.

The FDA also approved relugolix/estradiol/norethindrone combination tablets (Myfembree) for the treatment of moderate to severe endometriosis-associated pain. Relugolix is a GnRH receptor antagonist that binds to and blocks GnRH receptors in the anterior pituitary gland.

Approval was based on data from two trials (SPIRIT 1 and SPIRIT 2).[63] In both studies, 75% of women who used the once-daily relugolix combination therapy experienced a significant reduction in dysmenorrhea, compared with 27% and 30% of women in the placebo groups. The combination therapy also significantly reduced nonmenstrual pelvic pain in 59% and 66% of women in the treatment groups, compared with 40% and 43% in the placebo groups. Moreover, patients in the studies who received relugolix combination therapy reported decreases in dyspareunia and opioid use.

Add-back therapy

Estrogen/progestin "add-back" therapy can prevent bone mineral density loss and hypoestrogenic symptoms. Hormone therapy preparations, progestins, tibolone, and bisphosphonates have all been shown to be effective.[64,65] Add-back therapy has been demonstrated to prevent loss in bone density and to relieve vasomotor symptoms without reducing the efficacy of GnRH agonist regimens. GnRH agonists have been used for 12 months with norethindrone add-back therapy with good results.[66] The ESHRE recommends combined hormonal add-back therapy for patients receiving GnRH agonist therapy.[40]

GnRH analogues vs danazol

A Cochrane review comparing GnRH analogues with danazol treatment showed no difference in improvement of dysmenorrhea, dyspareunia, pelvic pain, or pelvic tenderness.[67] Likewise, no difference in retrospective American Fertility Society (rRAFS) score was found at approximately 24 weeks' follow-up. In contrast, studies that evaluated total pain resolution showed greater benefit from GnRH analogues compared with danazol. Adverse-effect profiles differed, with greater frequency of hot flushes and vaginal dryness with GnRH analogues, whereas danazol treatment resulted in a greater frequency of weight gain, acne, and headaches.

Danazol

Danazol acts by inhibiting the midcycle FSH and LH surges and preventing steroidogenesis in the corpus luteum.

Danazol has been shown to be as effective as any of the newer agents, but with a higher incidence of adverse effects. Its androgenic manifestations include oily skin, acne, weight gain, deepening of the voice, and facial hirsutism. Hypoestrogenic features due to danazol include emotional lability, hot flashes, vaginal dryness, and reversible breast atrophy.

In addition, because of the possibility of virilizing changes in a female fetus, the patient must use additional barrier contraception while receiving danazol therapy.

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Surgical Intervention

Surgical care can be broadly classified as follows:

- Conservative - When reproductive potential is retained
- Semiconservative - When reproductive ability is eliminated but ovarian function is retained
- Radical - When the uterus and ovaries are removed

Age, desire for future childbearing, and deterioration of quality of life are the main considerations when deciding on the extent of surgery.

Surgical efforts are aimed at removal of the endometrial implants and correction of anatomic distortions. Implants can be ablated using either laser energy or electrosurgical techniques.

Resection of the implants and adjacent peritoneum is considered the treatment of choice by some authors. A radical surgical approach involves total hysterectomy and bilateral salpingo-oophorectomy (TAH-BSO). This is generally reserved for women who have completed their family or for women with severely disabling pain that is unresponsive to more conservative approaches.

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Conservative Surgery

With conservative surgery, the aim is to destroy visible endometriotic implants and lyse peritubal and periovarian adhesions that are a source of pain and may interfere with ovum transport. The laparoscopic approach is the method of choice for treating endometriosis conservatively.

Ablation can be performed with laser or electrodiathermy. Overall, the recurrence rate is 19% and is similar for all techniques.[68] Laparoscopic ablative surgery with bipolar diathermy or laser for endometriomas was shown to be effective for relieving pelvic pain in 87% of patients.[69]

Drainage vs laparoscopic cystectomy

Ovarian endometriomas can be treated by drainage or cystectomy. Laparoscopic cystectomy was found to yield better pain relief and pregnancy rates than drainage.[70] Medical therapy with gonadotropin-releasing hormone (GnRH) agonists reduces the size of the cyst but does not influence pain relief.[71]

Ablation vs diagnostic laparoscopy

In the short term, the absolute benefit for women undergoing surgical ablation of endometriosis is 30-40% over women having only diagnostic laparoscopy. This benefit is reduced over time, and the reoperation rate is as high as 50%.[72] In cases of rectovaginal endometriosis, significant short-term pain relief was reported by up to 80%, but at 1-year follow-up, 50% required analgesics or hormones for pain relief. During postoperative treatment, GnRH analogues resulted in significantly reduced pain scores in women who received treatment for 6 months.

Presacral neurectomy and laparoscopic uterine nerve ablation

Presacral neurectomy is used to relieve severe dysmenorrhea. The nerve bundles are transected at the level of the third sacral vertebra, and the distal ends are ligated. Vascular injury to the middle sacral artery and vein is a potential complication, and some authors advocate prophylactic ligation. Also, constipation is a long-term adverse effect (94%) of this procedure and should be considered while deciding whether to perform this procedure.

Nodularity of the uterosacral ligaments may contribute to dyspareunia and low back pain. The transmission of neural pathways is via the Lee-Frankenhäuser plexus. Laparoscopic uterine nerve ablation (LUNA) is performed to interrupt the pain fibers. Potential complications of this procedure include uterine prolapse and pelvic denervation. A systematic review of trials of LUNA found no advantage in terms of pain relief when compared to placebo.[73] However, when combined with laparoscopic ablation, LUNA significantly reduced pain attributed to endometriosis.[74] In patients with subfertility, tissue ablation significantly increased the cumulative pregnancy rate.[50] A Cochrane review failed to show any benefit from either LUNA or presacral neurectomy.[75]

Postoperative adjunctive hormone therapy

For patients with endometriosis who are not immediately seeking pregnancy, postoperative adjunctive hormonal treatment is recommended to reduce the risk of recurrence of pain and endometriomas.[40] Combined oral contraceptives, progestogens (including the levonorgestrel-releasing intrauterine system), and GnRH agonists and antagonists are useful in this setting.

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Semiconservative Surgery

The indication for this semiconservative surgery is mainly in women who have completed their childbearing, are too young to undergo surgical menopause, and are debilitated by the symptoms. Such surgery involves hysterectomy and cytoreduction of pelvic endometriosis.

Ovarian endometriosis can be removed surgically, because one tenth of functioning ovarian tissue is all that is needed for hormone production. Patients who undergo hysterectomy with ovarian conservation have a 6-fold higher rate of recurrence compared to women who undergo oophorectomy.[76]

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Radical Surgery

Radical surgery involves total hysterectomy with bilateral oophorectomy (TAH-BSO) and cytoreduction of visible endometriosis. Adhesiolysis is performed to restore mobility and normal intrapelvic organ relationships.

Ureteric obstruction may warrant surgical release or excision of a damaged segment. Bowel obstruction may require a resection anastomosis or a wedge resection if the obstruction is confined to the anterior rectosigmoid.

Endometriosis may recur in 15% of women after extirpative surgery, irrespective of whether estrogen therapy is given postoperatively. Estrogen therapy can be instituted safely immediately after surgery, especially in younger women who face the prospect of accelerated bone loss and vasomotor symptoms.[77] No trials have reported the use of estrogen plus progestin therapy with respect to estrogen therapy alone postoperatively. However, theoretically, the addition of a continuous progestin could decrease the regrowth of endometriosis.

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Prevention

No current methods of prevention are known. Some evidence suggests that rapid and aggressive medical or surgical therapy can arrest progression, especially when the disease is caught in the early (minimal to mild) stages.

Early and prolonged use of oral contraceptive pills, pregnancy, and breastfeeding seem to afford some degree of protection against this disease.

A prospective cohort study by Farland et al found a 40% reduced risk of endometriosis for women who breastfed for 36 months or more throughout their reproductive lifetime compared with women who never breastfed. The investigators also reported an 8% lower risk of endometriosis for every additional 3 months of total breastfeeding per pregnancy.[78]

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Consultations

Treat patients with endometriosis in consultation with a physician experienced in the diagnosis and management of this condition and its complications, such as an obstetrician/gynecologist. If extensive disease is present, specialists in reproductive endocrinology, urology, colorectal surgery, and even gynecologic oncology may be required.

Any postpubertal patient undergoing surgery for acute or chronic pelvic/abdominal pain could have endometriosis; therefore, consultation with a physician having the experience to recognize, diagnose, and treat this disease is prudent. Conservation of future fertility may be dependent on the conservative and meticulous surgical approach of an expert reproductive surgeon.

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Long-Term Monitoring

If the presumptive diagnosis is endometriosis and follow-up is arranged, pain management can include the use of nonsteroidal anti-inflammatory drugs (NSAIDs) or narcotic analgesics. Start patients with classic symptoms of endometriosis and no reason to suspect another cause on medical therapy. Surgical diagnosis is not always required.

Lack of rapid response (within 1-2 cycles) to medical therapy should prompt a search for other causes of the patient's symptoms. Consider diagnostic laparoscopy if it has not been performed previously.

In patients who underwent surgery for endometriosis (or had endometriosis discovered during surgery for another indication), consider adjuvant medical treatment. At a minimum, prescribe oral contraceptive pills until these patients are ready to conceive.

Endometriosis is progressive and can result in chronic pain and infertility. Gynecologic follow-up is advised.

Pediatric patients

Although most pediatric patients are not currently interested in becoming pregnant, subsequent fertility is likely a major concern. Evidence is mounting that early and aggressive therapy may alter the course of this disease.[79] Investigate moderate to severe dysmenorrhea that is unresponsive to NSAIDs and pelvic pain persisting longer than 3 months.

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Guidelines

Guidelines Summary

The following organizations have released guidelines for the management of endometriosis. Key diagnostic and treatment recommendations have been reviewed and integrated throughout the article.

- 2024. National Institute for Health and Care Excellence. Endometriosis: diagnosis and management[39]
- 2022. European Society of Human Reproduction and Embryology. ESHRE guideline: endometriosis[40]
- 2018. American College of Obstetrics and Gynecology. ACOG committee opinion no. 760: dysmenorrhea and endometriosis in the adolescent[29]

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Medication

Oral Contraceptives

Desogestrel/ethinyl estradiol (Apri, Azurette, Caziant)

Norgestimate/ethinyl estradiol (Estarylla, Mili, Mono-Linyah)

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Progestational Agents

Medroxyprogesterone (Depo-SubQ Provera 104, DepoProvera, MPA)

Megestrol (Megace, Megace ES)

Norethindrone acetate (Aygestin, Norlutate)

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Gonadotropin-Releasing Hormone Analogues

Goserelin (Zoladex, Zoladex LA)

Leuprolide (Camcevi, Camcevi ETM, Eligard)

Nafarelin (Synarel)

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Gonadotropin-Releasing Hormone Antagonists

Elagolix (Orilissa)

Relugolix/estradiol/norethindrone (Myfembree)

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Antigonadotropic Agents

Danazol

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Aromatase Inhibitors

Letrozole (Femara)

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Questions & Answers

Overview

How is endometriosis defined?

When is radical surgery indicated in the treatment of endometriosis?

What are the signs and symptoms of endometriosis?

How is endometriosis diagnosed?

What are the most common sites of involvement in endometriosis?

When is histology indicated in the diagnosis of endometriosis?

What is the role of lab studies in the diagnosis of endometriosis?

What is the role of imaging studies in the diagnosis of endometriosis?

Which drugs are used in the treatment of endometriosis?

How is surgical care for endometriosis classified?

What is included in conservative surgical treatment for endometriosis?

When is semiconservative surgery indicated in the treatment of endometriosis?

How is endometriosis defined?

What is the typical location of ectopic implants in endometriosis?

What causes endometriosis?

What causes endometriosis?

What are the risk factors for endometriosis?

What is the role of retrograde menstruation in the etiology of endometriosis?

What is the role of immunologic dysfunction in the etiology of endometriosis?

What is the role of metaplasia in the etiology of endometriosis?

What is the role of remnant Mullerian cells in the etiology of endometriosis?

What is the role of genetics in the etiology of endometriosis?

What are the common routes of anatomic spread of endometriosis?

What are less common sites of anatomic spread of endometriosis?

What causes postmenopausal endometriosis?

What is the pathophysiology of endometriosis?

What is the prevalence of endometriosis?

Which women are at highest risk for endometriosis?

What is the prognosis of endometriosis?

Which treatments are effective for pain in endometriosis?

How does use of oral contraception affect the risk of developing endometriosis?

What are the advantages and disadvantages of surgical therapy for endometriosis?

How likely is pregnancy in patients with endometriosis?

What is the morbidity associated with endometriosis?

What are complications of endometriosis?

What is included in patient education for endometriosis?

Presentation

Which clinical history is characteristic of endometriosis?

How does the degree of pain correlate with the severity of endometriosis?

What are symptoms of endometriosis?

How does the presentation of endometriosis vary based on the area of involvement?

How is cyclic pain characterized in endometriosis?

What causes acute exacerbations of endometriosis?

What is the prevalence of secondary dysmenorrhea in endometriosis?

What are the symptoms of dyspareunia in endometriosis?

What are uncommon cyclic symptoms in endometriosis?

What causes partial or complete bowel obstruction in endometriosis?

Which physical findings are characteristic of endometriosis?

How is endometriosis staged?

DDX

What conditions should be included in the differential diagnoses of endometriosis?

What are the differential diagnoses for Endometriosis?

Workup

Which studies should be performed in the workup for endometriosis?

What is the role of a complete blood count (CBC) in the workup of endometriosis?

What is the role of urinalysis in the workup of endometriosis?

What is the role of a serum cancer antigen 125 (CA-125) test in the workup of endometriosis?

What is the sensitivity and specificity of autoantibodies against Thomsen-Friedenreich (T) antigen (Gal beta1-3GalNAc) in the diagnosis of endometriosis?

What is the role of ultrasonography in the workup of endometriosis?

What is the role of MRI in the workup of endometriosis?

What is the role of CT scanning in the workup of endometriosis?

What is the role of laparoscopy in the workup of endometriosis?

What are the most common sites of involvement for endometriosis?

Which histologic features are characteristic of endometriosis?

Treatment

What are treatment option for endometriosis?

What are the rates of recurrent pain with medical and surgical therapy for endometriosis?

How is subfertility managed in endometriosis?

How does endometriosis affect the outcomes of women utilizing assisted reproductive technology?

How does endometriosis affect the spontaneous abortion rate?

What is the role of hormonal therapy in the treatment of endometriosis?

How are combination oral contraceptive pills (COCPs) administered for the treatment of endometriosis?

How are progestational agents administered in the treatment of endometriosis?

How are gonadotropin-releasing hormone (GnRH) analogues administered for the treatment of endometriosis?

What is the role of estrogen or progestin add-back therapy in the treatment of endometriosis?

How do gonadotropin-releasing hormone (GnRH) analogues compare to danazol for the treatment of endometriosis?

What is the role of danazol for the treatment of endometriosis?

When is barrier contraception required during the treatment of endometriosis?

What is the role of surgery in the treatment of endometriosis?

What are the aims of conservative surgery for the treatment of endometriosis?

How does drainage compare to laparoscopic cystectomy in the treatment of endometriosis?

What is the efficacy of surgical ablation for the treatment of endometriosis?

What is the role of presacral neurectomy in the treatment of endometriosis?

What is the role of laparoscopic uterine nerve ablation (LUNA) in the treatment of endometriosis?

What is the efficacy of postoperative adjunctive hormone therapy for endometriosis?

When is semi-conservative surgery indicated in the treatment of endometriosis?

What is the role of radical surgery in the treatment of endometriosis?

How is endometriosis prevented?

What specialist consultations are recommended for the treatment of endometriosis?

What is included in long-term monitoring of patients with endometriosis?

What is included in long-term monitoring of pediatric patients with endometriosis?

Medications

Which medications are used in the treatment of endometriosis?

Which medications in the drug class Oral Contraceptives are used in the treatment of Endometriosis?

Which medications in the drug class Progestational Agents are used in the treatment of Endometriosis?

Which medications in the drug class Gonadotropin-Releasing Hormone Analogues are used in the treatment of Endometriosis?

Which medications in the drug class Gonadotropin-Releasing Hormone Antagonists are used in the treatment of Endometriosis?

Which medications in the drug class Antigonadotropic Agents are used in the treatment of Endometriosis?

Which medications in the drug class Aromatase Inhibitors are used in the treatment of Endometriosis?

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